Computational Logic in the Era of AI

Equality, and Sets as Predicates

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This lecture



1 **Equality**

$x =_S y$: no predicate P tells x and y apart

`Eq`, `Eq.subst`

► An equality lets you replace one side by the other anywhere.

2 **Motives**

$=$ is symmetric, transitive, a congruence

`rw`, `calc`, `congrArg`

► Each use of an equality picks a predicate; `rw` picks it for you.

3 **Definitional equality**

conversion ($\beta\delta$) against proof ($=_S$)

`rfl`, `simp`

► `rfl` closes what computes; the rest needs a proof.

4 **Uniqueness**

$\exists^1$, descriptor ι

`Classical.choose`

► A provably unique thing may be named, whatever the proof.

5 **Subsets**

$ps(S) := S \to *_p$, $x \varepsilon V$, $\subseteq$

`Subtype`, `Fin`

► A subset is a predicate, so membership is a proposition.

History: what it means to be equal



G. W. Leibniz

1646–1716

Equal means indiscernible: no predicate tells the two apart.

1686 Leibniz: things are the same when one may be substituted for the other everywhere, salva veritate.

1879 Frege's Begriffsschrift: the identity sign between names, with substitution as an axiom; later "the *F*", a definite description.

1908 Zermelo: a subset is cut out of a set by a predicate, the axiom of separation.

1972 Martin-Löf: equality as a type with one constructor and an elimination rule. Lean's Eq.



Gottlob Frege

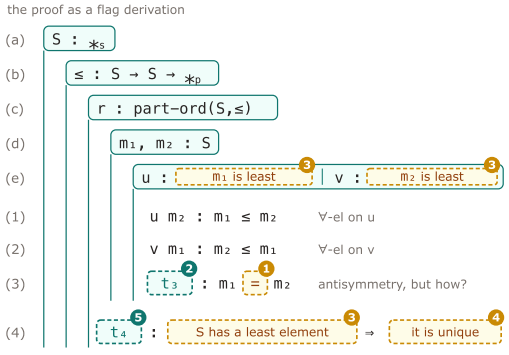
1848–1925

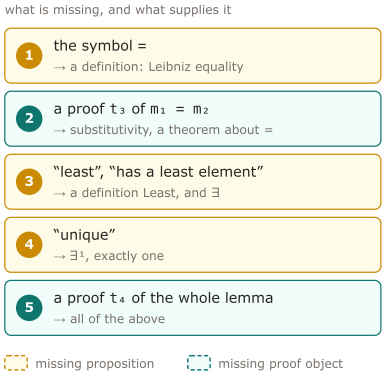
ι names the unique thing with a property.

Both ideas become definitions in this lecture: Leibniz's as $=_S$, Frege's as ι. Photos: Wikimedia. N&G §12.2, §12.7, §13.1

A first piece of mathematics

Lemma. If S is partially ordered by $\leq$ and has a least element, that element is unique.
Proof. Let m_1, m_2 be least: $m_1 \leq m_2$ and $m_2 \leq m_1$, so $m_1 = m_2$ by antisymmetry. □



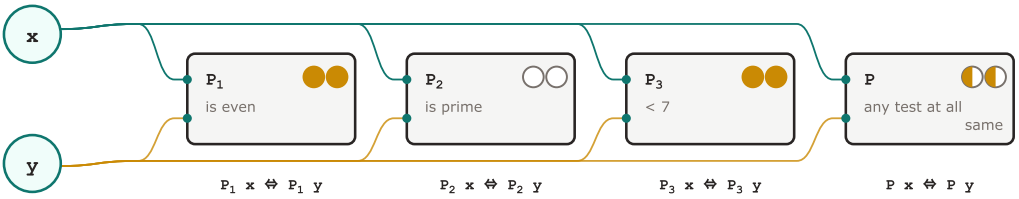


Each numbered gap is filled later; its slide carries the badge.

N&G §12.1, Fig. 12.1

What does $x = y$ mean?

Leibniz: x and y are equal when no predicate P tells them apart.

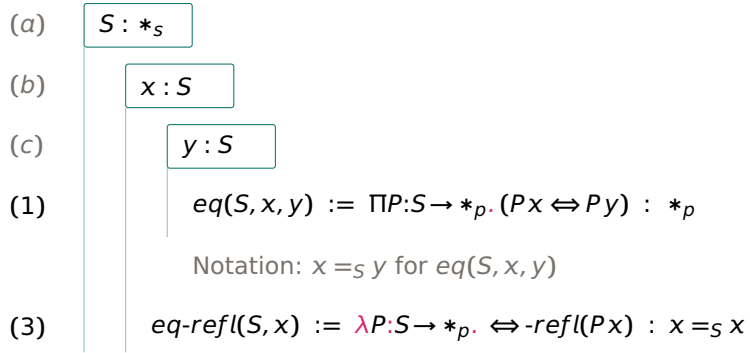


$\mathrm{eq}(S, x, y) := \Pi\, P : S \to *_p .\ (P\ x \Leftrightarrow P\ y)$ — indiscernible by every predicate

Second-order: the Π ranges over predicates, not elements.

Not an axiom but a definition, so the properties of equality are theorems. N&G §12.2

Equality as a definition



- Second order: P ranges over predicates, $S \to *_p : \Box$.
- Reflexivity is one λ. Even $\Rightarrow$ in place of $\Leftrightarrow$ would do.

N&G §12.2, Figs. 12.2–12.3, Rem. 12.2.1–12.2.2

Leibniz equality, written out

```
-- eq(S,x,y) := ΠP : S → *ₚ. (P x ⟺ P y)
-- "x and y are equal iff they are INDISCERNIBLE by any predicate."
def LEq {S : Type} (x y : S) : Prop := ∀ P : S → Prop, P x ↔ P y

-- reflexivity, by instantiating at P x ⟺ P x
theorem leq_refl {S : Type} (x : S) : LEq x x := fun _ => Iff.rfl

-- ⇒ instead of ⟺ would already do: P x ⇒ P y for all P gives the converse.
def LEq' {S : Type} (x y : S) : Prop := ∀ P : S → Prop, P x → P y

-- The Π ranges over PREDICATES, P : S → * : □.  So this is a
-- SECOND-ORDER definition, not a first-order ∀ over elements.
#check @LEq      -- @LEq : {S : Type} → S → S → Prop
```

Substitutivity: the rule for using an equality

Substitute equals for equals: if $x =_S y$ and Px, then Py, for every predicate P.

(a)	$S : *_s$	
(b)	$P : S \to *_p$	
(c)	$x, y : S$	
(d)	$u : x =_S y$	
(1)	$a_1 := uP : Px \Leftrightarrow Py$	∀-el on (d)
(2)	$a_2 := \Leftrightarrow\text{-}el_1(Px, Py, a_1) : Px \Rightarrow Py$	
(e)	$v : Px$	
(3)	$eq\text{-}subs(S, P, x, y, u, v) := a_2\, v : Py$	⇒-el

Using an equality means choosing P.

N&G §12.2, Fig. 12.4

What Lean uses instead

```
-- Lean does not use the coding.  'Eq' is an inductive with ONE
-- constructor, and its recursor is what substitutivity becomes.
#print Eq
-- inductive Eq.{u_1} : {α : Sort u_1} → α → α → Prop
-- number of parameters: 2
-- constructors: Eq.refl : ∀ {α : Sort u_1} (a : α), a = a

#check @Eq.subst
-- ∀ {α : Sort u_1} {motive : α → Prop} {a b : α}, a = b → motive a → motive b
-- Compare eq-subs(S,P,x,y,u,v) : P y.  Same shape; the predicate P is
-- what Lean calls the 'motive'.
```

The predicate *P* is what Lean calls the `motive`. Why `Eq.subst` is the recursor of Eq: a later lecture.

The two agree

```
-- The two notions coincide.  One direction needs only reflexivity ...
theorem eq_to_leq {S : Type} {x y : S} (h : x = y) : LEq x y :=
  fun P => h ▸ Iff.rfl

-- ... the other instantiates the quantifier at the predicate (x = ·),
-- the same trick that derives symmetry from substitutivity.
theorem leq_to_eq {S : Type} {x y : S} (h : LEq x y) : x = y :=
  (h (fun z => x = z)).mp rfl
```

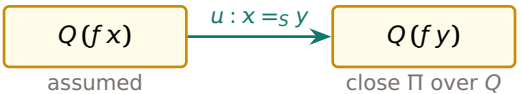
One direction is substitution with the motive $\lambda z.\,(x = z)$, and that motive is the whole proof.

Congruence, proved twice

Congruence. If $u : x =_S y$ and $f : S \to T$, then $f\,x =_T f\,y$. Two proofs, two choices of predicate:

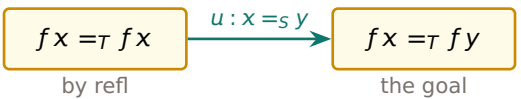
first proof

$\lambda z{:}S.\, Q\,(f\,z)$,
for any $Q : T \to *_p$



second proof

$Q_1 := \lambda z{:}S.\, (f\,x =_T f\,z)$

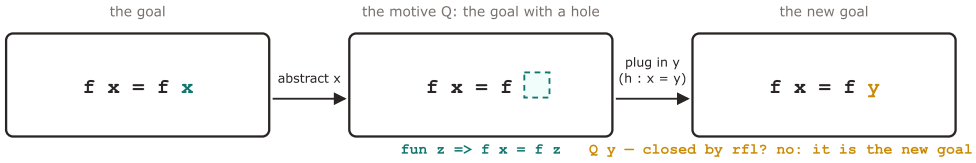


The second is one line: $\mathit{eq\text{-}cong}_2 := \mathit{eq\text{-}subs}(S, Q_1, x, y, u, \mathit{eq\text{-}refl}(T, f\,x))$.

N&G §12.3, Figs. 12.5–12.6

Choosing Q is choosing the motive

To use $x =_S y$, say which occurrences of x become the hole. That predicate is Lean's `motive`.



N&G §12.3, §12.5

“motive is not type correct”

```
-- ‘rw‘ builds the MOTIVE Q by abstracting ‘n‘ out of the goal, then checks
-- it.  Here the goal mentions ‘v : Fin (n+1)‘ (a natural below n+1, bundled
-- with that proof), whose TYPE mentions n -- so the abstraction is ill-typed:
#guard_msgs(drop error) in
example (n m : Nat) (v : Fin (n + 1)) (h : n = m) : v.val < n + 1 := by
  rw [h]
-- Tactic ‘rewrite‘ failed: motive is not type correct:
--   fun _a => ↑v < _a + 1
-- Error: Application type mismatch: The argument  v  has type
--   Fin (n + 1)  but is expected to have type  Fin (_a + 1) ...

-- Two cures: ‘subst‘ replaces n EVERYWHERE, v’s type included; ‘▸‘ takes
-- the motive from the proof term’s type instead of from the goal.
example (n m : Nat) (v : Fin (n + 1)) (h : n = m) : v.val < n + 1 := by
  subst h; exact v.isLt
example (n m : Nat) (v : Fin (n + 1)) (h : n = m) : v.val < n + 1 :=
  h ▸ v.isLt
```

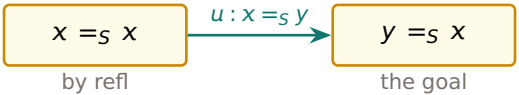
No predicate Q with $Q\,n$ the goal typechecks here: “choose Q” is a real error message.

Symmetry and transitivity, from substitutivity

Two more choices of predicate, with $u : x =_S y$ and $v : y =_S z$:

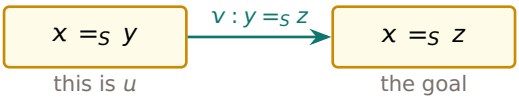
symmetry

$Q_2 := \lambda z{:}S.\ (z =_S x)$



transitivity

$Q_3 := \lambda w{:}S.\ (x =_S w)$



$eq\text{-}sym(S, x, y, u) := eq\text{-}subs(S, Q_2, x, y, u, eq\text{-}refl(S, x)) : y =_S x$

$eq\text{-}trans(S, x, y, z, u, v) := eq\text{-}subs(S, Q_3, y, z, v, u) : x =_S z$

$Q_2\, x =_\beta x =_S x$: Lean does ($conv$) silently.

N&G §12.5, Figs. 12.10–12.12

Motives in Lean

```
-- Congruence: the motive inferred from the goal, or given.
theorem cong' {S T : Type} (f : S → T) {x y : S} (h : x = y) : f x = f y :=
  h ▸ rfl                                          -- motive inferred
theorem cong'' {S T : Type} (f : S → T) {x y : S} (h : x = y) : f x = f y :=
  @Eq.subst S (fun z => f x = f z) x y h rfl       -- motive Q₁

-- Symmetry and transitivity, with Q₂ and Q₃ given.
theorem sym' {S : Type} {x y : S} (h : x = y) : y = x :=
  @Eq.subst S (fun z => z = x) x y h rfl           -- Q₂ x is x = x
theorem trans' {S : Type} {x y z : S} (h : x = y) (k : y = z) : x = z :=
  @Eq.subst S (fun w => x = w) y z k h             -- Q₃ y is x = y

-- In the library: ‘congrArg‘, ‘Eq.symm‘, ‘Eq.trans‘.
```

rw chooses the motive for you

```
-- ‘rw [h]‘ is eq-subs with the predicate chosen FOR you: it abstracts every
-- occurrence of h’s left side out of the goal, and that abstraction is Q.
theorem rw1 (a b : Nat) (h : a = b) (f : Nat → Nat) : f a + a = f b + b := by
  rw [h]           -- Q := fun z => f z + z = f b + b; the rest is Eq.refl

example (a b : Nat) (h : a = b) (k : b = 0) : a = 0 := by
  rw [h, k]        -- one equation after another, each left to right
example (a b : Nat) (h : b = a) : a = b := by
  rw [← h]         -- right to left
example (a b : Nat) (h : a = b) (k : a + 1 = 5) : b + 1 = 5 := by
  rw [h] at k; exact k                -- in a hypothesis

#print rw1
-- fun a b h f => Eq.mpr (id (congrArg (fun _a => f _a + _a = f b + b) h))
--                       (Eq.refl (f b + b))
```

Read the term: `congrArg` at the motive is congruence, `Eq.mpr` is substitution along it, and `Eq.refl` closes what is left. All of the equality theory so far is inside one `rw`.

Chains of steps: `calc`

```
-- ‘calc‘ chains equalities step by step.  Each step names its
-- justification; transitivity is applied for you.
def d (n : Nat) : Nat := n + n

example (a b : Nat) (h : a = b) (k : b = 5) : d a = 10 :=
  calc d a = d b := by rw [h]      -- congruence
    _      = d 5 := by rw [k]
    _      = 10  := rfl              -- definitional: δ, then arithmetic

-- Mixed relations chain too; the result is the strongest one used.
example (a b c d : Nat) (h₁ : a ≤ b) (h₂ : b < c) (h₃ : c ≤ d) : a < d :=
  calc a ≤ b := h₁
    _ < c := h₂
    _ ≤ d := h₃
```

Orders

✓ 3

"Partial order" is five definitions in one context:

(a) $S : *_s$

(b) $\leq : S \to S \to *_p$

Notation: $x \leq_S y$ for $\leq x\, y$

(1) $refl(S, \leq) := \forall x{:}S.\ (x \leq_S x) : *_p$

(2) $trans(S, \leq) := \forall x, y, z{:}S.\ (x \leq_S y \Rightarrow y \leq_S z \Rightarrow x \leq_S z) : *_p$

(3) $pre\text{-}ord(S, \leq) := refl(S, \leq) \wedge trans(S, \leq) : *_p$

(4) $antisymm(S, \leq) := \forall x, y{:}S.\ (x \leq_S y \Rightarrow y \leq_S x \Rightarrow x =_S y) : *_p$

(5) $part\text{-}ord(S, \leq) := pre\text{-}ord(S, \leq) \wedge antisymm(S, \leq) : *_p$

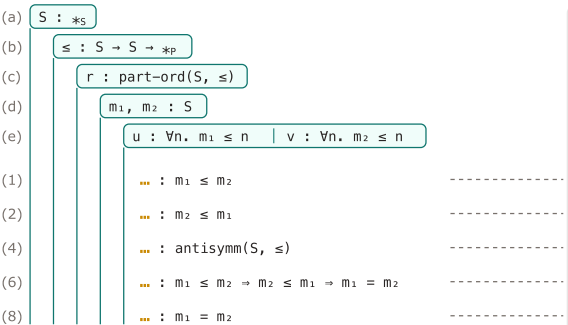
3 $Least(S, \leq, m) := \forall n{:}S.\ (m \leq_S n).$

N&G §12.4, Fig. 12.7, Rem. 12.4.1

A skeleton proof: the types, and nothing else

First the flags and the types of the lines; every proof is still a hole.

skeleton proof: only the types



the same skeleton, as a script

```
-- in context: the five flags (a)–(e)
-- S, le, r, m₁ m₂, u, v
have h₁ : m₁ ≤ m₂ := ?_
have h₂ : m₂ ≤ m₁ := ?_
have anti : Antisymm le := ?_
have step : m₁ ≤ m₂ → m₂ ≤ m₁ → m₁ = m₂ := ?_
show m₁ = m₂
```

the gold dots are the holes; the filled proof fills them, and `exact`/`rw` fill them in Lean

Filling the holes fills **2**.

N&G §12.4, Fig. 12.8

The proof objects, filled in

✓ 2

(*a*)	$S : *_s \mid \leq : S \to S \to *_p \mid r : \mathit{part\text{-}ord}(S, \leq)$	
(*d*)	$m_1, m_2 : S$	
(*e*)	$u : \forall n{:}S.\,(m_1 \leq_S n) \mid v : \forall n{:}S.\,(m_2 \leq_S n)$	
(1)	$a_1 := u\, m_2 \;:\; m_1 \leq_S m_2$	∀-el
(2)	$a_2 := v\, m_1 \;:\; m_2 \leq_S m_1$	∀-el
(4)	$a_4 := \wedge\text{-}el_2(\ldots, r) \;:\; \mathit{antisymm}(S, \leq)$	$\wedge$-el$_2$, then (*conv*)
(6)	$a_6 := a_4\, m_1\, m_2 \;:\; m_1 \leq_S m_2 \Rightarrow m_2 \leq_S m_1 \Rightarrow m_1 =_S m_2$	∀-el twice
(8)	$a_8 := a_6\, a_1\, a_2 \;:\; m_1 =_S m_2$	⇒-el twice
(9)	$a_9(S, \leq, r) := \lambda m_1 m_2{:}S.\,\lambda u.\,\lambda v.\,a_8 \;:\; \forall m_1, m_2{:}S.\,(\cdots \Rightarrow m_1 =_S m_2)$	(*abst*)

2 t_3 is a_8. Lines (3), (5), (7) only unfold, by (*conv*).

N&G §12.4, Fig. 12.9

The same proof, as named lines

```
-- The flag derivation, line for line.  A ‘have‘ is a named line; ‘exact‘
-- is the last.
theorem least_unique {S : Type} (le : S → S → Prop) (r : PartOrd le)
    (m₁ m₂ : S) (u : ∀ n, le m₁ n) (v : ∀ n, le m₂ n) : m₁ = m₂ := by
  have a₁ : le m₁ m₂ := u m₂                    -- (1)  ∀-el on u
  have a₂ : le m₂ m₁ := v m₁                    -- (2)  ∀-el on v
  have a₄ : Antisymm le := r.2                  -- (3)-(4)  ∧-el₂ on r
  have a₆ : le m₁ m₂ → le m₂ m₁ → m₁ = m₂ := a₄ m₁ m₂   -- (5)-(6)
  exact a₆ a₁ a₂                               -- (7)-(8)

#print least_unique
-- fun {S} le r m₁ m₂ u v => have a₁ := u m₂; have a₂ := v m₁;
--   have a₄ := r.right; have a₆ := a₄ m₁ m₂; a₆ a₁ a₂
```

The haves survive in the term as local definitions: the flag derivation, folded into one object.

Two equalities

propositional: `x =S y`, a proposition; an inhabitant is needed (`Eq`, `rw`, `calc`)

definitional: $M =_{\beta\delta\iota} N$, a judgement;
the kernel decides it (`rfl`)

- `2 + 2 = 4`
- `d 3 = 6` (δ, then arithmetic)
- `n + 0 = n` (ι on the second argument)

- `0 + n = n`
  (needs induction)
- `a + b = b + a`
  (needs a lemma)
- `V = W`
  (needs an axiom: extensionality)

everything inside the teal region is also in the gold one, by `rfl`; nothing goes the other way

Teal sits inside gold, not the other way round: that is why `rw` exists. N&G §4.7, §9.7, §12.2

Definitional and propositional, side by side

	definitional	**propositional**
what it is	a judgement	a proposition
written	$M =_{\beta\delta} N$	$M =_S N$
comes from	the ($conv$) rule	a definition
decided by	the kernel	nobody: it needs a proof
closed by	`rfl`	`rw`, `Eq.subst`, `calc`

$n + 0 = n$ is definitional: addition recurses on its second argument. $0 + n = n$ is not: it needs an induction.

N&G §4.7, §12.2

Where `rfl` stops, and what takes over

```
-- ‘Nat.add‘ recurses on its SECOND argument, so one of these computes and
-- the other is stuck on the variable n:
example (n : Nat) : n + 0 = n := rfl
#guard_msgs(drop error) in
example (n : Nat) : 0 + n = n := rfl
-- The type of ‘Eq.refl ?a‘ ... is not definitionally equal to ‘0 + n = n‘

-- What closes it is a theorem from Δ -- proved by induction, next lecture:
example (n : Nat) : 0 + n = n := Nat.zero_add n

-- With n closed, computation is back, and ‘decide‘ runs it in the kernel:
example : 0 + 5 = 5 := by decide
```

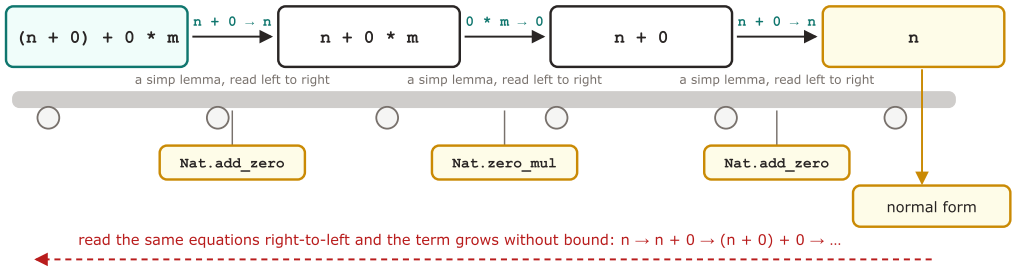
Past the boundary: a lemma, a closed instance the kernel runs, or rewriting to a normal form.

Rewriting to a normal form

A rewrite rule is an equation read one way, $s \to t$. `simp` rewrites with every lemma tagged `@[simp]`: its simp set.

Orientation is the whole design decision.

confluence and termination are theorems for βδι; for your simp set they are your obligation.



N&G §1.9

Unique existence

4 Three grades of “exists”, for $P : S \to *_p$:

	definition	**reads**	**e.g.**
$\exists^{\geq 1}(S, P)$	$\exists x{:}S.\, P\,x$	at least one	○ ● ○ ● ●
$\exists^{\leq 1}(S, P)$	$\forall y, z{:}S.\, (P\,y \Rightarrow P\,z \Rightarrow y =_S z)$	at most one	○ ○ ○ ○ ○ ○ ○ ● ○ ○
$\exists^{1}(S, P)$	$\exists^{\geq 1}(S, P) \wedge \exists^{\leq 1}(S, P)$	exactly one	○ ○ ● ○ ○

● an element with P

5 Line (9) of the proof is $\exists^{\leq 1}$ for $P := \mathit{Least}$; with $w : \exists^{\geq 1}$, $\wedge$-in gives $\exists^{1}$.

Theorems and proofs

If $\Gamma \vdash a(\bar{x}) := N : *_p$ and $\Gamma \vdash b(\bar{x}) := M : N$, then $a(\bar{x})$ is a theorem and $b(\bar{x})$ its proof.

N&G §12.6, Figs. 12.14–12.15

The descriptor ι

What is unique can be named: $\iota_{x:S}(Px)$, the x with Px. Two primitive definitions, the object and its property:

(a)	$S : *_s \mid P : S \to *_p \mid u : \exists^1 x{:}S.\, Px$	
(1)	$\iota(S,P,u) := \perp\!\!\!\perp : S$	no body
	Notation: $\iota^u_{x:S}(Px)$ for $\iota(S,P,u)$	
(2)	$\iota\text{-}prop(S,P,u) := \perp\!\!\!\perp : P(\iota^u_{x:S}(Px))$	no body

- Whatever has P is the ι: $\forall z{:}S.\, (Pz \Rightarrow z =_S \iota^u_{x:S}(Px))$.
- So the proof u does not matter: $\iota(S,P,u_1) =_S \iota(S,P,u_2)$.
- The minimum: $Min := \iota(S, \lambda m.\, Least(S,\leq,m), \ldots)$.

N&G §12.7, Figs. 12.16–12.18, Rem. 12.7.2

`Classical.choose` is ι

```
-- ∃¹ is "at least one" and "at most one".  Core Lean has no ∃¹, so the
-- definition is written out.
def ExUnique {S : Type} (P : S → Prop) : Prop :=
  (∃ x, P x) ∧ ∀ y z, P y → P z → y = z

-- ι(S,P,u) := ⊥⊥ : S  and  ι-prop(S,P,u) := ⊥⊥ : P (ι ...).
-- Lean's are 'Classical.choose' and 'choose_spec', primitive too:
noncomputable def iota {S : Type} (P : S → Prop) (u : ExUnique P) : S :=
  Classical.choose u.1
theorem iota_prop {S : Type} (P : S → Prop) (u : ExUnique P) : P (iota P u) :=
  Classical.choose_spec u.1
#print axioms iota_prop     -- 'iota_prop' depends on axioms: [Classical.choice]

-- Whatever satisfies P is the ι.
theorem eq_iota {S : Type} (P : S → Prop) (u : ExUnique P) (z : S) (v : P z) :
    z = iota P u := u.2 z _ v (iota_prop P u)
```

`noncomputable`: a primitive constant has no body to run, and the axiom audit names it.

Irrelevance of proof

```
-- λD does not make proof irrelevance a principle.  Lean's kernel has it
-- for Prop, definitionally: two proofs of one proposition are rfl-equal.
example (p : Prop) (h₁ h₂ : p) : h₁ = h₂ := rfl

-- So the worry that ι(S,P,u₁) and ι(S,P,u₂) might differ
-- cannot even be stated as a difference:
example {S : Type} (P : S → Prop) (u₁ u₂ : ExUnique P) :
    iota P u₁ = iota P u₂ := rfl

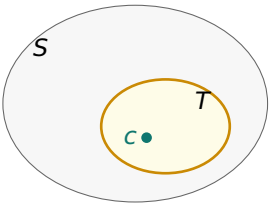
-- What keeps it consistent: a proof may not be turned into DATA.  A Prop's
-- recursor eliminates only into Prop, so no function can tell two
-- proofs apart:
#guard_msgs(drop error) in
def theWitness (h : ∃ n : Nat, 3 < n) : Nat := match h with | ⟨n, _⟩ => n
-- error: recursor 'Exists.casesOn' can only eliminate into 'Prop'
```

λD proves the equation; Lean's kernel makes it a conversion. That is one of the places Lean is not *λD*.

Why subsets cannot be types

Suppose a subset $T \subseteq S$ were a type, so that $c : T$.

one term, two types



$c : S$ and $c : T$:
Uniqueness of Types fails

type checking becomes proof search

type-check $F\,M$, with
F the square root

needs $M : \{x \in \mathbb{R} \mid x \geq 0\}$

find a proof of $M \geq 0$

undecidable in general
(Church, Turing 1936)

So typing stays decidable, and subsets need another representation. N&G §13.1

Subsets are predicates

A subset of S is a predicate on S; predicates are already in λD.

(a)	$S : *_s$
(1)	$ps(S) := S \to *_p : \Box$
(b)	$x : S \mid V : ps(S)$
(2)	$element(S, x, V) := V\,x : *_p$ Notation: $x \,\varepsilon\, V$
(c)	$V, W : ps(S)$
(3)	$\subseteq (S, V, W) := \forall x{:}S.\,(x \,\varepsilon\, V \Rightarrow x \,\varepsilon\, W) : *_p$
(4)	$\cup(S, V, W) := \lambda x{:}S.\,(x \,\varepsilon\, V \vee x \,\varepsilon\, W) : ps(S)$

$x \,\varepsilon\, V$ is a proposition to prove, not a typing $x : V$. N&G §13.2, Figs. 13.1–13.2

ps(*S*), built rather than imported

```
-- The subset definitions, transcribed in ‘Clai.Sets‘ (Lean CORE has no
-- ‘Set‘: that is Mathlib).  Thirty-odd lines, nothing hidden.
#check @PS     -- Type → Type                        (1) ps(S) := S → *_p
#check @Mem    -- {S : Type} → PS S → S → Prop       (2) element(S,x,V) := V x
#check @Subs   -- {S : Type} → PS S → PS S → Prop    (3) ⊆(S,V,W)
example (S : Type) (V W : PS S) :
    (V ⊑ W) = ∀ x : S, x ∈ V → x ∈ W := rfl      -- the guarded ∀, verbatim

-- On paper V ⊆ W; here V ⊑ W.  ‘⊆‘ is core’s ‘HasSubset‘ notation; a
-- symbol of our own keeps definition (3) visibly THE definition, and
-- later lectures read the same symbol as entailment.
```

Membership needs a proof

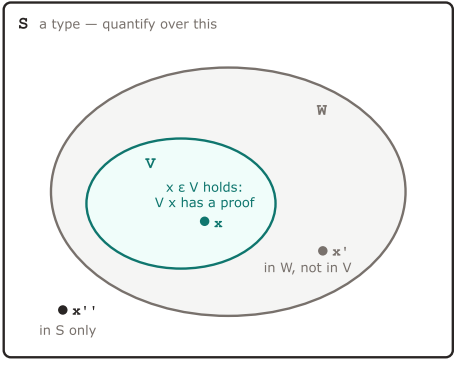
```
-- x ∈ V is not x : V but x ε V, which is a TYPE.  To establish x ∈ V
-- we have to give a PROOF.
def Evens : PS Nat := fun n => n % 2 = 0

-- ‘4 ∈ Evens‘ unfolds to ‘4 % 2 = 0‘, so a computation proves it.
example : 4 ∈ Evens := show 4 % 2 = 0 by decide
#guard_msgs(drop error) in
example : 5 ∈ Evens := show 5 % 2 = 0 by decide   -- fails: 5 % 2 = 1

-- The rules ε-in and ε-el are both the identity: ‘x ε {x | P x}‘
-- is β-equal to ‘P x‘, so nothing needs doing.
example (n : Nat) (h : n % 2 = 0) : n ∈ Evens := h
```

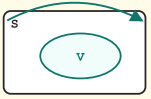
Every quantifier ranges over a type

V is not a type, so $\forall x : V$ is not well formed. Quantify over S and guard:



V ⊆ W := ∀ x : S, x ε V ⇒ x ε W

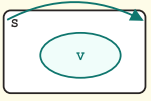
∀ x : S, (x ε V ⇒ P x) — the guarded ∀



x sweeps all of S

outside V the guard is vacuous;
inside V it demands P x. ✓

∀ x : S, (x ε V ∧ P x) — too strong



x sweeps all of S

asserts every x of S lies in V
— false unless V = S. ✗

∀ x ε V. … is not well formed:
V is a predicate S → $*_p$, not a type.
∃ guards with ∧, not ⇒.

The existential guards with ∧: $\exists x{:}S.\,(x \,\varepsilon\, V \wedge P\,x)$. N&G §13.2, Ex. 13.2.1

Guarded quantifiers

```
-- The translation convention.  Every quantifier ranges over a TYPE;
-- the subset becomes a GUARD:
--   ∀x∈V (P x)  ⇝  ∀x : S. (x ε V ⇒ P x)      -- implication
--   ∃x∈V (P x)  ⇝  ∃x : S. (x ε V ∧ P x)      -- CONJUNCTION
example (V : PS Nat) (P : Nat → Prop) : Prop := ∀ x, x ∈ V → P x
example (V : PS Nat) (P : Nat → Prop) : Prop := ∃ x, x ∈ V ∧ P x

-- The asymmetry is not a typo.  With ∧ in the universal case, the claim
-- would assert that every x is in V; with ⇒ in the existential case, it
-- would be satisfied by any x outside V.
```

A proof about sets

For all subsets V and W of S, if $V \subseteq W^c$ then $V \setminus W = V$.

(a)	$S : *_s \mid V, W : ps(S)$	
(b)	$\quad x : S \mid v : x \,\varepsilon\, V \setminus W$	
(2)	$\quad a_2 := \ldots$ use $\wedge$-el$_1$ on v $\ldots$: $x \,\varepsilon\, V$	$v : x \,\varepsilon\, V \wedge \neg(x \,\varepsilon\, W)$, by $(conv)$
(4)	$a_4(S, V, W) := \ldots$ use $\Rightarrow$-in and $\forall$-in $\ldots$: $V \setminus W \subseteq V$	
(c)	$\quad u : V \subseteq W^c$	
(d)	$\quad\quad x : S \mid v : x \,\varepsilon\, V$	
(5)	$\quad\quad a_5 := u\, x\, v$: $x \,\varepsilon\, W^c$	$\forall$-el, $\Rightarrow$-el
(8)	$\quad\quad a_8 := \ldots$ use $\wedge$-in on v and a_5 $\ldots$: $x \,\varepsilon\, V \setminus W$	$x \,\varepsilon\, W^c =_\beta \neg(x \,\varepsilon\, W)$
(10)	$\quad a_{10}(S, V, W, u) := \ldots$: $V \subseteq V \setminus W$	
(12)	$a_{12}(S, V, W) := \ldots$ use $\wedge$-in on a_4 and a_{10}, then $\Rightarrow$-in $\ldots$: $(V \subseteq W^c) \Rightarrow (V \setminus W = V)$	

Each “use $\ldots$” names one rule. Only a_{10} depends on u.

N&G §13.2, Fig. 13.4

The same proof, as a script

```
-- The flag derivation, as a script.  ‘intro x v‘ raises the flags x : S
-- and v; the hints "use ∧-el₁", "use ∧-in", "use ⇒-in and ∀-in" become
-- ‘.1‘, ‘⟨_, _⟩‘ and ‘intro‘.
theorem diff_eq {S : Type} (V W : PS S) (u : V ⊆ compl W) :
    IS (diff V W) V := by
  constructor
  · intro x v                          -- lines (1)-(4): no use of u
    exact v.1                          -- (2) ∧-el₁ on v : x ε V ∧ ¬(x ε W)
  · intro x v                          -- lines (5)-(10)
    have a₅ : x ∈ compl W := u x v      -- (5) u x v : x ε W^c
    exact ⟨v, a₅⟩                      -- (7)-(8) ∧-in: x ε V\W

-- The second half needs u and the first does not, and the parameter
-- lists show it (a₄(S,V,W) against a₁₀(S,V,W,u)).
```

A specification is a predicate, that is, a subset

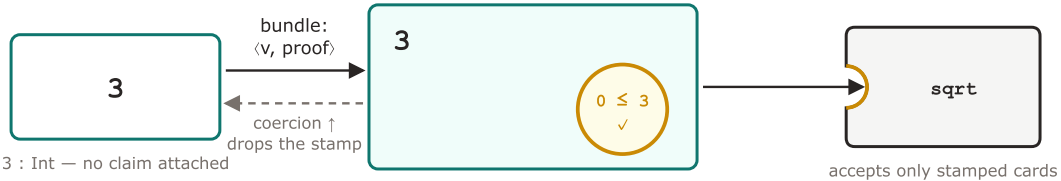
a specification, in words	**as a predicate**
the sorted lists	$\lambda xs{:}List\,\mathbb{N}.\ sorted(xs)\ :\ ps(List\,\mathbb{N})$
the inputs on which f returns 0	$\lambda n{:}\mathbb{N}.\ (f\,n =_S 0)\ :\ ps(\mathbb{N})$
the states from which a loop terminates	$\lambda s{:}State.\ \ldots\ :\ ps(State)$
an assertion about a program state	$\lambda s{:}State.\ (0 < s)\ :\ ps(State)$
"P entails Q", P the stronger	$P \subseteq Q$
a predicate transformer	a map $ps(State) \to ps(State)$

"The program satisfies it" is a membership $x\,\varepsilon\,V$, to be proved. Nothing is added to the type theory to state a specification.

N&G §13.2

The other route: carry the proof

A subtype value is the element with its proof stamped on.



subsets-as-TYPES were weighed and rejected:
checking M : {x | x ≥ 0} means FINDING a proof — undecidable.
A subtype keeps typing decidable: the proof is an explicit component, checked, never searched.

The proof travels with the element: checked, never searched for.

Subtypes in Lean

```
-- Bundle the element WITH its proof:
-- the proof is carried, not searched for.
def NonNeg := {n : Int // 0 ≤ n}

def three : NonNeg := ⟨3, by decide⟩
#eval three.val        -- 3
#check three.property
-- three.property : 0 ≤ three.val

-- The cost: ‘3 : Int‘ and ‘three : NonNeg‘
-- are different terms of different types;
-- a coercion moves between them.
-- ‘Fin n‘ is ‘{i : Nat // i < n}‘.
```

Extensionality is an axiom

Same elements ($V \subseteq W \wedge W \subseteq V$) does not give Leibniz equality $V = W$ without an axiom.

```
-- Lean's counterparts of the axiom: 'funext' and 'propext'.
theorem setext {α : Type} {V W : PS α} (h : ∀ x, x ∈ V ↔ x ∈ W) : V = W :=
  funext (fun x => propext (h x))

#print axioms setext
-- 'setext' depends on axioms: [propext, Quot.sound]
-- TWO, not one.  'propext' turns the ↔ into an =; but 'funext' is NOT
-- primitive in Lean -- it is derived from quotient soundness, so
-- 'Quot.sound' appears too.  λD needed an axiom; so does Lean.
```

One axiom in λD, two in Lean. N&G §13.2, Fig. 13.5

What is still missing: numbers

Equality, orders and sets are definitions. Numbers are not: in λD they are six primitive definitions, and they do not compute.

The Peano axioms in λD

(1)–(3)	$\mathbb{N} := \bot\!\!\!\bot : *_s, \quad 0 := \bot\!\!\!\bot : \mathbb{N}, \quad s := \bot\!\!\!\bot : \mathbb{N}\to\mathbb{N}$	the set, zero, successor
(4)	$ax\text{-}nat_1 := \bot\!\!\!\bot : \forall x{:}\mathbb{N}.\ \neg(s\,x =_{\mathbb{N}} 0)$	
(5)	$ax\text{-}nat_2 := \bot\!\!\!\bot : \forall x, y{:}\mathbb{N}.\ (s\,x =_{\mathbb{N}} s\,y \Rightarrow x =_{\mathbb{N}} y)$	
(6)	$ax\text{-}nat_3(P) := \bot\!\!\!\bot : (P\,0 \wedge \forall x{:}\mathbb{N}.\ (P\,x \Rightarrow P\,(s\,x))) \Rightarrow \forall x{:}\mathbb{N}.\ P\,x$	induction

And without recursive definitions, fac is not a program: it cannot be executed, so $fac(3) = 6$ requires a proof.

Next lecture

Inductive types: two constructors in place of six axioms, and $fac\,3 = 6$ by `rfl`.

N&G §8.2, §14.1

References I

Avigad, J., L. de Moura, S. Kong, and S. Ullrich (2026). *Theorem Proving in Lean 4*. With contributions from the Lean community; tested against Lean 4.33.0. URL: https://lean-lang.org/theorem_proving_in_lean4/ (visited on 08/20/2026).

Frege, G. (1893). *Grundgesetze der Arithmetik*. Jena: Pohle.

Nederpelt, R. and H. Geuvers (2014a). "Mathematics in λD: a first attempt". In: *Type Theory and Formal Proof: An Introduction*. Cambridge: Cambridge University Press. Chap. 12, pp. 257–278.

— (2014b). "Numbers and arithmetic in λD". In: *Type Theory and Formal Proof: An Introduction*. Cambridge: Cambridge University Press. Chap. 14, pp. 305–348.

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— (2014d). *Type Theory and Formal Proof: An Introduction*. Cambridge: Cambridge University Press.

References II

N&G chapters 12 and 13 throughout. Ch. 12: §12.1 (Lemma 12.1.2, Fig. 12.1), §12.2 (Figs. 12.2–12.4, Rem. 12.2.1–12.2.2), §12.3 (Figs. 12.5–12.6), §12.4 (Fig. 12.7, Rem. 12.4.1, Figs. 12.8–12.9), §12.5 (Figs. 12.10–12.12), §12.6 (Figs. 12.13–12.15, p. 271), §12.7 (Figs. 12.16–12.18, Lemma 12.7.1, Rem. 12.7.2). Ch. 13: §13.1, §13.2 (Figs. 13.1–13.5, Ex. 13.2.1, Rem. 13.2.2), §13.3 (Fig. 13.6). §14.13 for irrelevance of proof; §10.5 for the confluence and normalisation results quoted on the rewriting slide. The Lean slides follow *Theorem Proving in Lean 4*: §4.1–4.3 (equality, `calc`), §5.6–5.7 (`rw`, `simp`), ch. 11 (`conv`) and §12.2–12.5 (`propext`, `funext`, choice).